

ESP32-S3 Water Potability & Treatment

Technical Knowledge Base & RAG Grounding Manual | Version 3.0

Operational Purpose: Comprehensive specification document establishing parameter thresholds, deterministic potability evaluation logic, root-cause diagnostics, and automated treatment pipeline architectures.

1. Primary Parameter Thresholds & Evaluation Logic

Water telemetry is continuously evaluated against three primary physical and chemical parameters: Potential of Hydrogen (pH), Total Dissolved Solids (TDS), and Electrical Conductivity (EC). To classify as **POTABLE**, water telemetry must satisfy ALL conditions simultaneously. Any single parameter violation immediately flags the water as NOT POTABLE.

pH Classification (1.0 - 14.0)

- **Potable (Ideal):** 6.5 to 8.5 (Balanced neutral state)
- **Acidic Range:** 1.0 to 6.49 (Corrosive risk)
- **Alkaline Range:** 8.51 to 14.0 (Scaling/caustic risk)

TDS (mg/L) & EC (µS/cm)

- **TDS Ideal:** 50.0 to 300.0 mg/L
- **TDS Acceptable:** 300.1 to 500.0 mg/L
- **EC Ideal:** 0.0 to 400.0 µS/cm
- **EC Acceptable:** 400.1 to 800.0 µS/cm

2. Multi-Parameter Conflict Resolution

High TDS + Low pH (e.g., pH 5.4, TDS 850) Cause: The water contains high dissolved solids that are acidic in nature. Action: Install a reverse osmosis unit followed by an alkaline post-filter.
High TDS + High pH (e.g., pH 9.2, TDS 1100) Cause: The water has high TDS composed of alkaline scaling minerals. Action: Install a cation softener followed by a reverse osmosis unit.
Normal TDS + Extreme pH (e.g., pH 3.5, TDS 120) Cause: The water suffers from severe chemical acid contamination. Action: Install a sodium carbonate dosing pump to normalize the pH.

3. Root Cause Analysis Matrix: pH Deviations (Acidic)

Granular diagnostic evaluation for acidic water anomalies. Action plans are prioritized sequentially.

Violation Category	Range	Expanded Cause Analysis	System Recommendation & Action Plan
Extremely Acidic	1.0 - 1.99	Severely acidic, causing dangerous heavy metal leaching from pipes. Highly toxic and corrosive to standard plumbing materials.	1. Install an industrial-grade sodium carbonate dosing pump to neutralize the acid. 2. Shut down main water lines immediately to halt toxic flow. 3. Restrict all usage until baseline pH levels are fully restored.
Highly Acidic	2.0 - 2.99	Exhibits high corrosivity, actively dissolving copper and lead fixtures. Ingestion poses serious health risks due to dissolved metals.	1. Install a high-capacity calcite and sodium carbonate neutralizing tank. 2. Flush all plumbing lines thoroughly once the system is active. 3. Clear residual metallic contaminants from the water supply.
Strongly Acidic	3.0 - 3.99	Severe risk of structural damage to pipes. Carries a distinct metallic, sour taste and is completely unfit for consumption.	1. Deploy a sodium carbonate dosing pump paired with an RO unit. 2. Monitor sensor telemetry daily to track automated injection. 3. Ensure the automated pump maintains a strictly neutral pH output.
Acidic	4.0 - 4.99	Noticeable sour taste; accelerates gradual wear on metal pipes. Continuous use leads to pinhole leaks in household plumbing.	1. Pass the water supply through a dual-tank limestone bed. 2. Periodically backwash the limestone bed to prevent channeling. 3. Maintain efficient treatment to elevate alkalinity naturally.
Mildly Acidic	5.0 - 5.99	Introduces metallic flavor and slow, progressive plumbing corrosion. Prolonged exposure causes blue-green copper staining.	1. Install a standard calcite media filter to naturally raise the pH. 2. Replenish the sacrificial calcite media annually as it dissolves. 3. Monitor for blue-green copper staining on sinks and fixtures.
Slightly Acidic	6.0 - 6.49	Sits just below the ideal limit. Mildly corrosive to sensitive plumbing. Requires minor balancing to achieve full potability.	1. Install a simple inline remineralization cartridge to buffer the pH. 2. Use this cartridge alongside a standard carbon block filter. 3. Improve overall drinking taste and stabilize the supply.

4. Root Cause Analysis Matrix: pH Deviations (Alkaline)

Advanced diagnostic tables for alkaline scaling conditions. Part 1 of 2.

Violation Category	Range	Expanded Cause Analysis	System Recommendation & Action Plan
Slightly Alkaline	8.51 - 8.99	Elevated bicarbonate levels significantly reduce chlorine effectiveness. Water feels slippery and leaves mild scale on glass.	1. Install a cation exchange resin filter to reduce excess alkalinity. 2. Ensure regular sanitization of the entire water system. 3. Counteract the lowered natural disinfection capabilities.
Alkaline	9.0 - 9.99	Produces a bitter taste and actively causes hard mineral scale buildup in pipes. Appliance heating efficiency is heavily reduced.	1. Install a weak acid softener system to lower the high pH. 2. Clean internal scale from existing appliances once treated. 3. Use a dedicated descaling solution for thermal appliances.
Strongly Alkaline	10.0 - 10.99	Causes heavy scaling blockages and presents a high pH chemical hazard. Prolonged contact causes severe skin dryness and irritation.	1. Install an automated HCl (hydrochloric acid) dosing pump. 2. Implement strict automated pH monitoring in the pipeline. 3. Prevent any accidental acid overdosing in the water system.
Highly Alkaline	11.0 - 11.99	Officially classified as a chemical hazard, carrying severe risks of tissue damage. Cannot be safely used for domestic washing.	1. Install an aggressive acid dosing system followed by an RO unit. 2. Use extreme caution if handling raw water leaks. 3. Wear appropriate personal protective equipment during maintenance.

4.1 Root Cause Analysis Matrix: pH Deviations (Severe Alkaline)

Advanced diagnostic tables for extreme alkaline scaling conditions. Part 2 of 2.

Violation Category	Range	Expanded Cause Analysis	System Recommendation & Action Plan
Extremely Alkaline	12.0 - 12.99	Highly caustic, entirely toxic to consume, and damaging to biological matter. Rapid pipe degradation is highly likely.	1. Implement an industrial-grade hydrochloric acid dosing loop. 2. Restrict all physical access to this water source immediately. 3. Wait until a full professional remediation is complete.
Dangerously Caustic	13.0 - 13.99	Chemically equivalent to bleach. Causes immediate skin irritation and severe burns. Vapors pose extreme safety emergencies.	1. Shut the main supply valve immediately to halt the water flow. 2. Dose the standing water with neutralizing acid only. 3. Maintain strict safety protocols and

5. TDS & EC Evaluation Rules

Guidelines managing mineral concentrations and electrical conductivity telemetry.

Violation Category	Range	Expanded Cause Analysis	System Recommendation & Action Plan
Very Low Minerals	TDS < 50 EC < 75	Lacks essential mineral ions, resulting in a flat taste. Overly pure water becomes "hungry" and aggressively leaches metals.	1. Install a dedicated calcium and magnesium remineralization cartridge. 2. Replace the remineralization cartridge every 6-12 months. 3. Prevent overly pure water from leaching metals from pipes.
High Minerals	TDS 500-1000 EC 800-1500	Distinctly hard water, leading to noticeable scale buildup and cloudiness. Alters food taste and reduces soap lathering.	1. Install a 100GPD reverse osmosis (RO) membrane system. 2. Pair the RO unit with a carbon pre-filter block. 3. Protect the sensitive RO membrane from premature fouling.
Severe TDS/EC	TDS > 1000 EC > 1500	Classified as brackish; entirely unsafe for human consumption. Rapidly corrodes standard household plumbing and appliances.	1. Install an industrial-grade water distiller or multi-pass RO system. 2. Secure alternative freshwater sources for primary consumption. 3. Avoid processing this highly brackish water if treatment is too costly.

6. Granular TDS Quality & Taste Stratification

- **0.0-49.9 ppm:** Flat taste. *Action: Add remineralizer.*
- **50.0-150.0 ppm:** Ideal. *Action: None.*
- **150.1-300.0 ppm:** Tap quality. *Action: None.*
- **300.1-500.0 ppm:** Hard limit. *Action: Carbon block.*
- **500.1-1000.0 ppm:** Salty/scaling. *Action: RO system.*
- **1000.1-2000.0 ppm:** Brackish. *Action: Dual-pass RO.*
- **> 2000.0 ppm:** Saline (toxic). *Action: Thermal distiller.*

7. Temperature Compensation

- **Baseline Temperature:** 25°C
- **Formula:** $EC_{25} = EC_{\text{measure}} / (1 + 0.02 \times (T_{\text{measured}} - 25))$
- **Formula:** $TDS = EC_{25} \times 0.65$

8. Sensor Diagnostics & Telemetry Faults

Fault 1 (Dry Probe):

pH ≤ 0.5 AND EC = 0.0

"SENSOR FAULT: Probes are dry."

Fault 2 (Logic Conflict):

TDS > 0.0 BUT EC = 0.0

"CALIBRATION FAULT: Check EC wiring."

Fault 3 (Out of Bounds):

pH < 0.0 OR pH > 14.0

"CALIBRATION FAULT: Recalibrate pH."

9. Standardized Treatment Pipeline Architectures

Pipeline A (pH < 6.5):